Summary in Graph

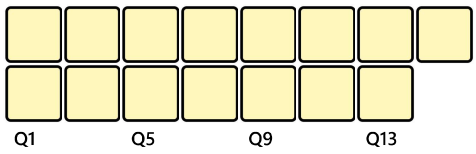
Exam Summary (GO Classes DA Test Series 2026 | Probability and Statistics | Topic Wise Test 4)

Qs. Attempted:	**0** (0 + 0)	**Correct Marks:**	**0** (0 + 0)
Correct Attempts:	**0** (0 + 0)	**Penalty Marks:**	**0** (0 + 0)
Incorrect Attempts:	**0** (0 + 0)	**Resultant Marks:**	**0** (0 + 0)

Total Questions:	**15** (10 + 5)
Total Marks:	**20** (10 + 10)
Exam Duration:	**45 Minutes**
Time Taken:	**0 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (62) out of 618 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Express the probability $P(\{-1 < X \leq 4\} \cap \{2 < Y \leq 5\})$ in terms of $F_{X,Y}(x, y)$, the joint CDF of the random variables X and Y.

A. $F_{X,Y}(4,5) - F_{X,Y}(-1,5) - F_{X,Y}(4,2) - F_{X,Y}(-1,2)$
B. $F_{X,Y}(4,5) + F_{X,Y}(-1,5) + F_{X,Y}(4,2) + F_{X,Y}(-1,2)$
C. $F_{X,Y}(4,5) - F_{X,Y}(-1,5) + F_{X,Y}(4,2) - F_{X,Y}(-1,2)$

D. $F_{X,Y}(4,5) - F_{X,Y}(-1,5) - F_{X,Y}(4,2) + F_{X,Y}(-1,2)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $Y = X + 30$ and $Z = 3X - 4$. X is a uniform random variable on the interval $[-1, 2]$.
Find $\mathrm{COV}(Y, Z)$.

A. $\frac{3}{4}$

B. $\frac{9}{4}$

C. $\frac{15}{4}$

D. $\frac{27}{4}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Consider three random variables, X, Y, and Z, with means $E(X) = 1$, $E(Y) = 2$, and $E(Z) = -1$. Given that X and Y are uncorrelated, and $E(XZ) = 5$, what is $\mathrm{COV}(X, Y + 2Z)$?

A. 12

B. -12

C. 2

D. -2

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $Z = 3X - Y - 5$, where X and Y are independent random variables with $\mathrm{Var}(X) = 1$ and $\mathrm{Var}(Y) = 2$. What is $\mathrm{Var}(Z)$?

A. 4

B. 7

C. 11

D. 16

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

If X is a uniformly distributed RV on the interval $[0, 1]$, then the pdf of the RV $Y = 5X$ is

A.

$$f_Y(y) = \begin{cases} 1/5 & \text{for } 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

B.

$$f_Y(y) = \begin{cases} 5 & \text{for } 0 \le y \le 1/5 \\ 0 & \text{otherwise} \end{cases}$$

C.

$$f_Y(y) = \begin{cases} 1/5 & \text{for } 0 \le y \le 5 \\ 0 & \text{otherwise} \end{cases}$$

D.

$$f_Y(y) = \begin{cases} 5 & \text{for } 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let a and b be some constant. Which of the following statement is false?

A. $\text{Cov}(a + bX, a + bY) = b^2 E[XY] - b^2 E[X]E[Y]$.
B. $\text{Var}(aX - bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) - 2ab \text{Cov}(X, Y)$.
C. $\text{Cov}(W, Z) = \text{Corr}(X, Y)$, where $W = (X - E(X))/\sqrt{\text{Var}(X)}$ and $Z = (Y - E(Y))/\sqrt{\text{Var}(Y)}$.
D. $\text{Cov}(X, W) = \text{Var}(X)$, where $W = (X - E(X))/\sqrt{\text{Var}(X)}$.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be a random variable. Define a new random variable Z as $Z = \frac{X - E(X)}{\sqrt{\text{Var}(X)}}$.
Given this definition, what property does the standardized random variable Z satisfy?

A. $E(Z) = 0$ and $\text{Var}[Z] = \text{Var}(X)$
B. $E(Z) = 1$ and $\text{Var}[Z] = 0$
C. $E(Z) = 0$ and $\text{Var}[Z] = 1$
D. $E(Z) = 1$ and $\text{Var}[Z] = 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

If the distribution of X is symmetric and reaches its highest point at the exact middle of the distribution:

A. The mode of X will be equal to the median of X.
B. The mean of X will be equal to the median of X.
C. The median of X will be at the exact middle of the distribution.
D. All of the above are true.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 09sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Which of the following statements is not true?

A. Two random variables can have the same mean but different medians.
B. Two random variables can have the same mean but different variances.
C. The mean of the sum of two random variables is always equal to the sum of their means.
D. The variance of the sum of two random variables is always equal to the sum of their variances.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Doubling the value of the largest observation in a sample of incomes will: (Mark all that are TRUE)

A. Increase the median.
B. Increase the mode.
C. Increase the mean.
D. decrease the median.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Imagine you're rolling a fair six-sided die repeatedly. If you get a 6, you win right away and the game stops. But if you roll any number from 1 to 5 , you have to wait for that many minutes before rolling again. For example, if you roll a 3 , you wait for 3 minutes before your next roll. What is the expected elapsed time from when you start rolling until you win? (Note: If you win on your first roll, the elapsed time is zero.)

A. 15 minutes

B. 10 minutes

C. 20 minutes

D. 25 minutes

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X be a random variable whose distribution depends on two conditions: C_1 and C_2, where C_1 and C_2 form a partition of the sample space, meaning $P(C_1) + P(C_2) = 1$. Given that $P(C_1) = \frac{1}{4}$ and the conditional cumulative distribution functions (CDFs) are

$$F_X(x \mid C_1) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{x}{2} & \text{for } 0 \leq x \leq 2 \\ 1 & \text{for } x > 2 \end{cases}$$

and

$$F_X(x \mid C_2) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{x}{3} & \text{for } 0 \le x \le 3 \\ 1 & \text{for } x > 3 \end{cases}$$

Find the probability $P(X \le 2)$.

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{2}{3}$

D. $\frac{3}{4}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Which of the following statements is/are TRUE?

A. If X and Y have $\mathrm{COV}(X, Y) = c_{XY}$, then $\mathrm{COV}(X - E(X), Y - E(Y)) = c_{XY} - E(X)E(Y)$.
B. $E(X) = \sum_{i=1}^{n} E(X \mid A_i)$ if $A_1, A_2, \ldots, A_n$ form a partition.
C. If $E(XY) \neq E(X)E(Y)$, then X and Y cannot be independent random variables.
D. $\mathrm{COV}(3X + 4, -2Y - 1) = 36\,\mathrm{COV}(X, Y)$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider a constant b. Let X be a random variable.
Given the expression $E\left[(X - b)^2\right]$, where $E[\cdot]$ denotes the expected value, what value of b minimizes $E\left[(X - b)^2\right]$?

A. $b = \sqrt{E[X]}$
B. $b = -\sqrt{E[X]}$
C. $b = E[X]$
D. $b = -E[X]$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Given the joint density function $f(x, y)$ of random variables X and Y :

$$f(x, y) = \begin{cases} cxy & \text{if } 0 \le y \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Which of the following is $E[Y \mid X = x]$, the conditional expectation of Y given $X = x$?

A. $\frac{x}{3}$

B. $\frac{x}{2}$

C. $\frac{2x}{3}$

D. $\frac{3x}{2}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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